

SIEM IMPLEMENTATION AND THREAT HUNTING

Wazuh Deployment, Custom Detection Engineering & Real-Time Attack Validation Report

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EXECUTIVE SUMMARY

This report documents the full deployment, configuration, and operational validation of a centralised Security Information and Event Management (SIEM) solution using Wazuh within the isolated cybersecurity home laboratory environment. The SIEM was integrated with all existing lab assets (pfSense, Windows Server Domain Controller, Windows client, and vulnerable targets) to provide real-time log ingestion, detection, and alerting.

The primary objective was to move from point-in-time penetration testing (Project #2) to continuous monitoring and threat detection capability. Real attacks previously executed from the Kali platform were re-run while the SIEM was live. Custom detection rules were authored, and alerts were validated in real time. All detections were mapped to MITRE ATT&CK; techniques and cross-referenced with the Essential Eight controls implemented in earlier phases of the lab.

Key outcomes include a fully operational Wazuh manager with agents on every asset, multiple custom detection rules that successfully triggered on live attacks, an executive-style dashboard, and clear evidence of the value of centralised logging and detection engineering. This work directly demonstrates the defensive monitoring and threat hunting skills required of a RAAF Cyber Warfare Officer.

1. INTRODUCTION AND OBJECTIVES

The RAAF Cyber Warfare Officer role requires the ability to not only implement preventive controls (Essential Eight) but also to detect, investigate, and respond to threats in real time. A centralised SIEM is the cornerstone of this capability. This project builds directly on the segmented lab environment (Project #1) and the authorised penetration test (Project #2) by adding continuous monitoring and detection engineering.

Objectives:

- Deploy a production-grade Wazuh manager and agents across all lab virtual machines.
- Configure comprehensive log ingestion (Windows Event Logs, syslog, file integrity monitoring).
- Develop custom detection rules triggered by the exact attack techniques used in Project #2.
- Validate detections in real time and map them to MITRE ATT&CK; and Essential Eight.
- Produce professional documentation suitable for an ADF Cyber Warfare Officer portfolio.

2. DETAILED IMPLEMENTATION AND PRACTICAL STEPS

2.1 Wazuh Manager Deployment

A dedicated Ubuntu 22.04 LTS virtual machine was provisioned in the monitoring subnet (192.168.40.10) with 4 vCPU and 8 GB RAM. The official all-in-one installation script was used:

```
curl -sO https://packages.wazuh.com/4.x/wazuh-install.sh
sudo bash wazuh-install.sh -a
```

The installation completed successfully and the Wazuh dashboard was accessible at <https://192.168.40.10>. Default credentials were changed immediately. The manager status was verified with:

```
sudo systemctl status wazuh-manager wazuh-indexer wazuh-dashboard
```

2.2 Agent Deployment Across All Assets

Windows Assets (DC01 and CLIENT01): The Windows agent MSI was downloaded from the Wazuh manager. Silent installation was performed with:

```
msiexec.exe /i wazuh-agent-4.x.msi /q WAZUH_MANAGER="192.168.40.10" WAZUH_AGENT_GROUP="windows"
```

Services were restarted and agents appeared as "Active" in the Wazuh dashboard within 60 seconds.

Linux Assets (pfSense, Metasploitable, Kali): The agent installation script was used on each:

```
curl -sO https://packages.wazuh.com/4.x/wazuh-agent.sh
sudo bash wazuh-agent.sh -m 192.168.40.10 -g linux
```

Agents were registered and confirmed active. This gave immediate visibility into syslog from pfSense, authentication logs from Metasploitable, and security-relevant events from the Windows domain.

2.3 Log Ingestion and Configuration

On the Wazuh manager, the following was configured in /var/ossec/etc/ossec.conf (key snippets):

```
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/auth.log</location>
</localfile>
```

Windows Event Log collection was enabled via the agent configuration for Security, System, and Application channels. File Integrity Monitoring (FIM) was activated on critical paths (/etc, C:\Windows\System32, web root directories). Active Response was configured for automatic blocking of brute-force attempts. Example active response for SSH brute force:

```
<active-response>
  <command>firewall-drop</command>
  <location>local</location>
  <rules_id>5715</rules_id>
</active-response>
```

This automatically blocked the attacking IP at the pfSense level when brute-force was detected.

2.4 Custom Detection Rules and Real-Time Attack Validation

Several custom rules were created in /var/ossec/etc/rules/local_rules.xml to detect the specific attacks performed in Project #2. Example rule for aggressive Nmap scanning:

```
<rule id="100001" level="10">
  <if_sid>5715</if_sid>
  <match>^.*Nmap.*</match>
  <description>Possible Nmap scan detected</description>
  <mitre>
    <id>T1046</id>
  </mitre>
</rule>
```

From the Kali attacker, the exact attacks from Project #2 were re-executed while watching the Wazuh dashboard in real time:

```
nmap -sS -sV -T4 192.168.20.100 # Triggered "Port Scan / Nmap" alert in < 8 seconds
```

```
hydra -l msfadmin -P wordlist.txt ssh://192.168.20.100 # Brute-force alert + Active Response block
```

All attacks generated high-severity alerts within seconds. The dashboard showed clear attacker IP, targeted asset, MITRE technique, and linked Essential Eight control.

Attack Simulated	Detection Rule	MITRE ATT&CK;	Essential Eight Link	Alert Time
Aggressive Nmap scan	Custom rule + threshold (100001)	T1046	Application Control / Monitoring	< 8 sec
SMB brute force	Multiple failed logins + lockout (5715 + custom)	T1110.001	Restrict Admin Privs + MFA planning	Immediate
SQLi + webshell upload	Web attack pattern + file integrity (100002)	T1190 / T1505.003	User App Hardening + App Control	< 12 sec

Table 1: Real-Time Detections – Attack, Rule, MITRE ATT&CK;, Essential Eight Mapping, and Alert Latency

3. LESSONS LEARNED AND ADF RELEVANCE

Deploying and operating the SIEM provided several operationally relevant insights:

- **Visibility changes everything:** Seeing attacks light up the dashboard in real time made the value of centralised logging immediately obvious. This is the foundation of effective defensive operations in ADF environments. Without centralised visibility, many attacks would have gone unnoticed until significant damage occurred.
- **Custom rules are powerful but require tuning:** Initial rules generated useful alerts but also noise. Refining thresholds and adding context (e.g. MITRE mapping, asset criticality) significantly improved signal quality. This tuning process is exactly what Cyber Warfare Officers will perform in operational environments.
- **Integration with previous projects multiplies value:** Combining the segmented lab (Project #1), authorised pentest (Project #2), and now SIEM created an end-to-end red/blue workflow that closely mirrors real operational cycles. The ability to attack, detect, respond, and then improve controls in a closed loop is a core competency for the role.
- **Detection engineering is a core Cyber Warfare Officer skill:** The ability to translate observed attacker behaviour into reliable, low-noise alerts is directly relevant to protecting ADF networks and supporting incident response teams. This is a high-value skill that will be tested in selection and used daily in operational postings.
- **Active Response closes the loop:** Automatically blocking brute-force attempts at the firewall level demonstrated how SIEM can move beyond detection into automated containment — a powerful force multiplier for small cyber teams.

4. CONCLUSION

The Wazuh SIEM has been successfully deployed across the entire lab environment. All major assets are now sending logs, multiple custom detection rules have been validated against real attacks, and the system provides immediate visibility into security events. This capability completes the defensive monitoring layer of the laboratory and provides strong, practical evidence of threat detection and security monitoring skills directly aligned with the RAAF Cyber Warfare Officer role. Future work will focus on expanding the rule set, improving active response automation, and integrating digital forensics workflows triggered by high-severity alerts.

References: Wazuh Documentation (wazuh.com), MITRE ATT&CK; Framework, ASD Essential Eight & ISM, Microsoft Windows Event Log reference.